

Clinical Risk Assessment Report: Patient 19

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Positive for Metastasis
- Prediction Confidence: Borderline / Low Confidence
- Absolute Predicted Risk: 25.0% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 1.13x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as High-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Patient Age (Age), which elevated the patient's absolute risk by +4.0%. Biologically, this feature patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.

Clinical Sensitivity Analysis (Borderline Patient):

PREDICTION INSTABILITY WARNING: This patient's risk profile sits precisely on the borderline. Consequently, the model's prediction is highly sensitive. Minor biological variations or measurement errors in the following features would flip the final clinical prediction:

- Patient Age: An increase would alter the prediction outcome.
- Small Zone Emphasis (PET): A decrease would alter the prediction outcome.
- Squamous Cell Carcinoma Antigen: An increase would alter the prediction outcome.
- 75th Percentile Tumor Density: A decrease would alter the prediction outcome.

2. Key Pathological & Imaging Drivers (Elevating Risk)

Patient Age (Age) **Value: 46.0 [Bottom 5%] (+4.0%)**

Patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.

Small Zone Emphasis (PET) (GLZLM_SZE.PET) **Value: 0.16 [Avg (56th %ile)] (+2.5%)**

Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

Squamous Cell Carcinoma Antigen (SCC_Ag) **Value: 0.6 [Bottom 18%] (+2.4%)**

Serum levels reflect tumor proliferation, burden, and squamous cell differentiation.

75th Percentile Tumor Density (CONVENTIONAL_HUQ3) **Value: 0.68 [Avg (68th %ile)] (+2.3%)**

Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.

Maximum Tumor Density (CONVENTIONAL_HUmax) **Value: 0.65 [Top 19%] (+1.8%)**

Reflects the most dense solid components or calcifications within the primary tumor lesion.

3. Protective / Mitigating Factors (Reducing Risk)

Minimum Tumor Density (CONVENTIONAL_HUmin) **Value: -0.56 [Avg (27th %ile)] (-2.4%)**

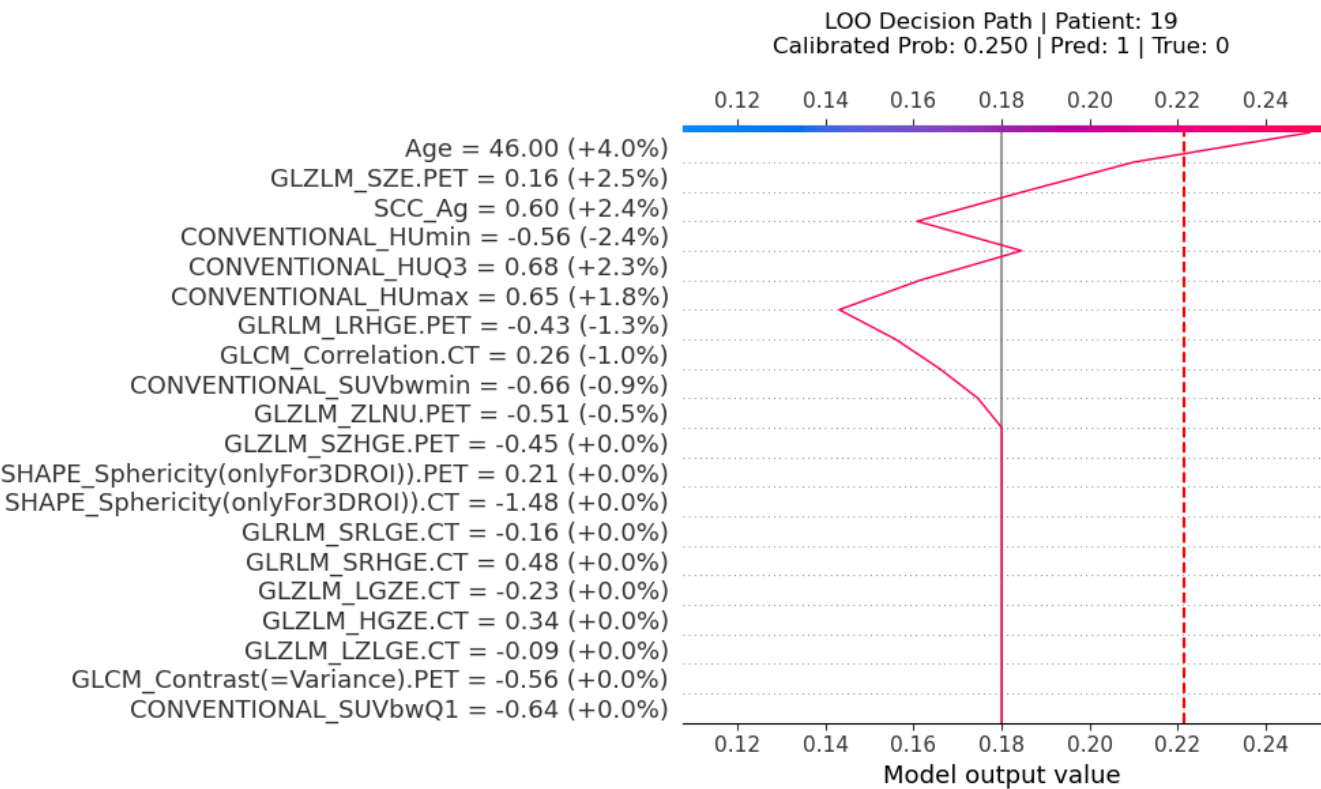
Reflects the least dense areas within the tumor on CT, often corresponding to cavitation, necrosis, or cystic degeneration.

Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET) **Value: -0.43 [Avg (39th %ile)] (-1.3%)**

Points to continuous, elongated bands of high-density or highly active tumor tissue.

Tumor Pixel Correlation (CT) (GLCM_Correlation.CT)**Value: 0.26 [Avg (57th %ile)] (-1.0%)**
Measures the linear dependency of gray levels; atypical values corroborate a complex, non-uniform spatial cellular arrangement.

Machine Learning Feature Attribution (SHAP Decision Path)



Sensitivity Analysis: Feature Tipping Points

The plots below demonstrate how variations in specific clinical or radiomic features affect the model's predicted risk. The red marker indicates the patient's current clinical state. Crossing the red dashed line changes the diagnosis.

